



TITM
UNIVERSITY

Process Intelligence
TITM University
2025/2026

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MASTER THESIS PROCESS INTELLIGENCE DASHBOARD

From process mining insights to management action
action

WHY THIS PROCESS MATTERS

The thesis process directly affects students, services and the university.



STUDENT EXPERIENCE

Repeated corrections and delays reduce confidence in the process.



ADMINISTRATIVE WORKLOAD

Rework increases pressure on Academic Services and the Library.



GOVERNANCE & TRANSPARENCY

Management needs fact-based visibility into where issues occur.



Submission



Review



Jury



Defense



Archiving

The goal was to replace internal perceptions with process evidence.

PROJECT SCOPE AND DATA BASIS

A complete view of 500 thesis cases.

500

THESIS CASES

9,698

EVENTS

24

ACTIVITIES

41

PROCESS VARIANTS

The event log records the real activity sequence from thesis submission to final notification. The case table adds context such as supervisor, programme, nationality, gender, age and final grade.

CASE TABLE



Context data



EVENT LOG



Activity sequences
with timestamps



CELONIS DASHBOARD



Management insights

THE AS-IS REALITY

Only one quarter of theses follow the ideal process path.

23.8%

HAPPY PATH
CONFORMANCE

119

HAPPY PATH CASES

381

CASES WITH
DEVIATIONS

41

PROCESS VARIANTS



24%

Happy path

76%

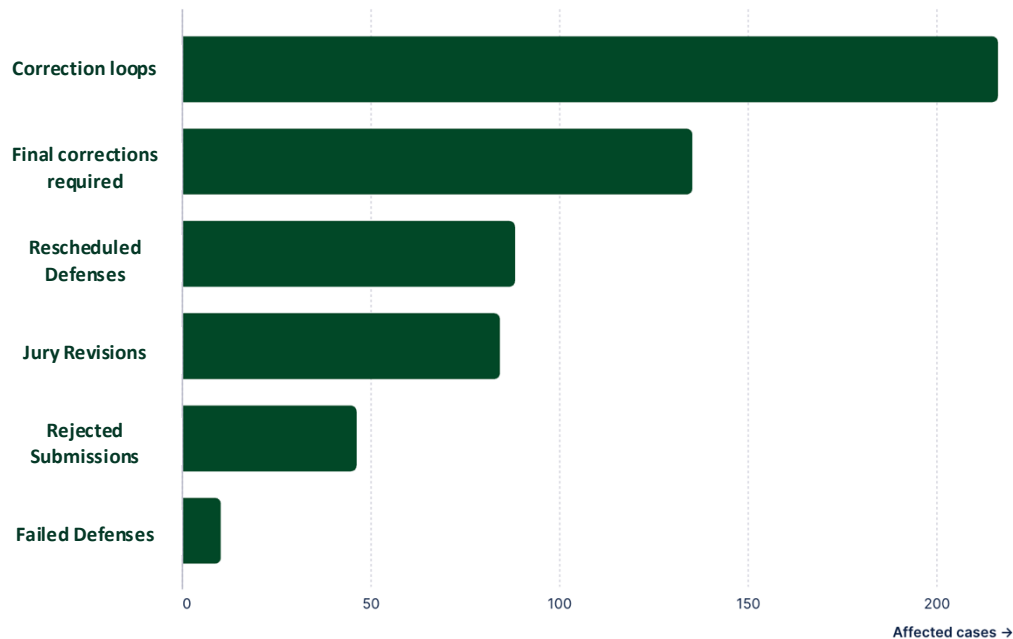
contain deviations

The process is structured on paper, but highly variable in practice.

WHERE THE PROCESS BREAKS

Correction loops are the largest source of non-conformance.

Main Deviation Drivers by Cases



216

theses required correction loops before the process could continue.

The largest issue is rework, followed by post-defense corrections and scheduling-related instability.

SCHOOL DIRECTOR HYPOTHESES

The data clarified six management hypotheses.

H1

**Supervisor
correction variation**



CONFIRMED

H2

**Defense
rescheduling drivers**



PARTIALLY REJECTED

H3

**Informatics
correction rate**



REJECTED

H4

**Correction loop
persistence**



WEAKLY CONFIRMED

H5

**Demographic
performance effect**



REJECTED

H6

**Supervision
practice effect**



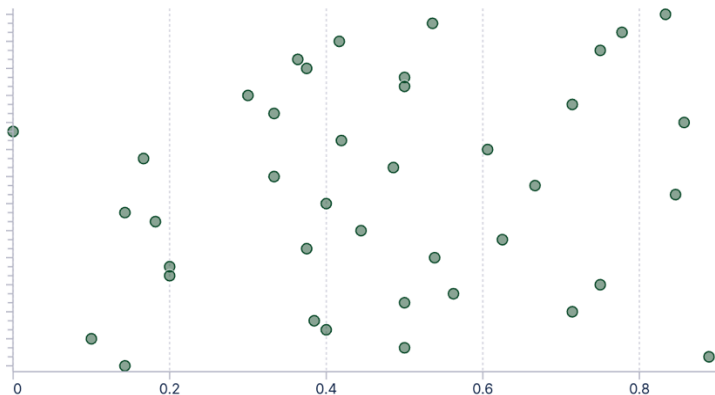
CONFIRMED

The strongest evidence points to process consistency and supervision practice, not student performance.

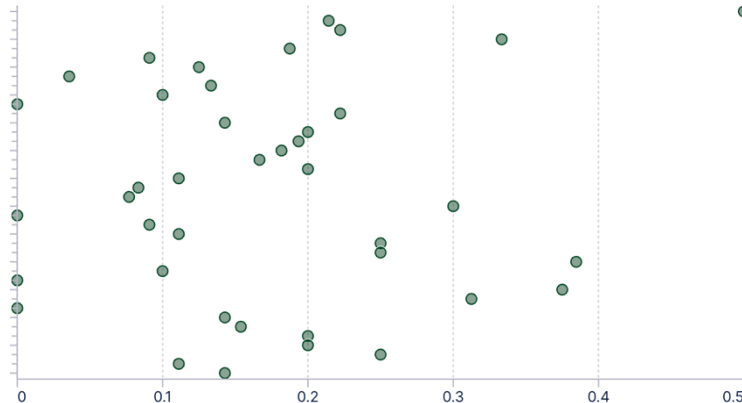
HYPOTHESIS EVIDENCE 1: SUPERVISION MATTERS

H1 confirmed | H6 confirmed

Average correction loops by supervisor



Average reschedules by supervisor



KEY INSIGHTS

Correction loops and reschedules vary meaningfully by supervisor.

Issues cluster around certain supervisors.

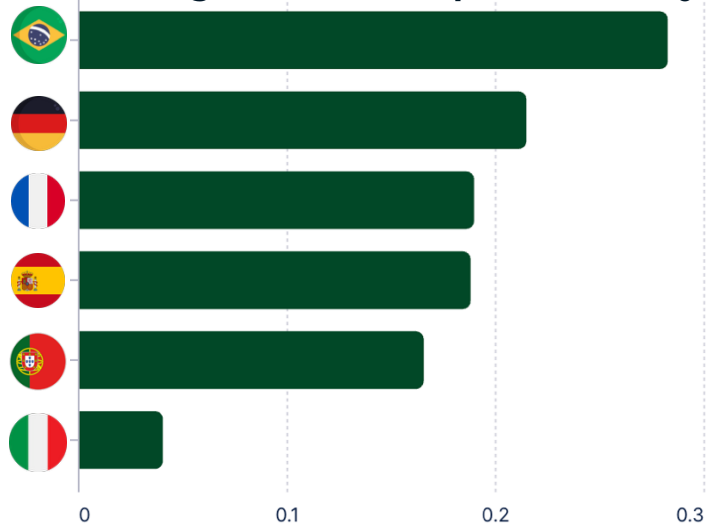
Final grades remain relatively stable across supervisors.

This suggests variation in supervision practice and process control, not simply weaker students.

HYPOTHESIS EVIDENCE 2: SCHEDULING & REWORK

H2 partially rejected | H4 weakly confirmed

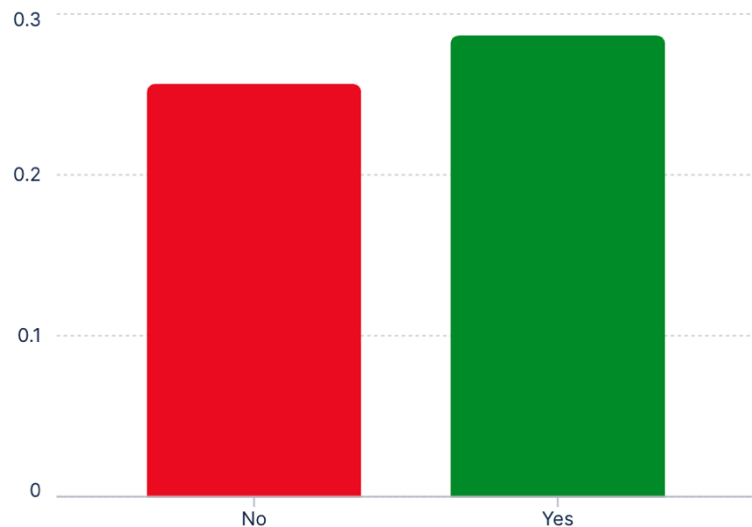
Average reschedules per nationality



H2: Partially Rejected

Rescheduling rates differ across student groups, so jury availability is likely not the only driver.

Final corrections by initial correction status



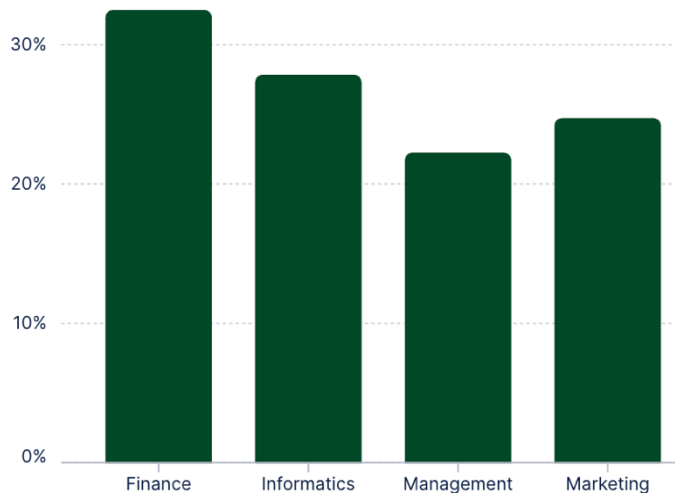
H4: Weakly Confirmed

Cases with initial corrections show slightly higher final correction risk, but the effect is weak.

HYPOTHESIS EVIDENCE 3: REJECTED ASSUMPTIONS

H3 rejected | H5 rejected

Post-defense corrections by master

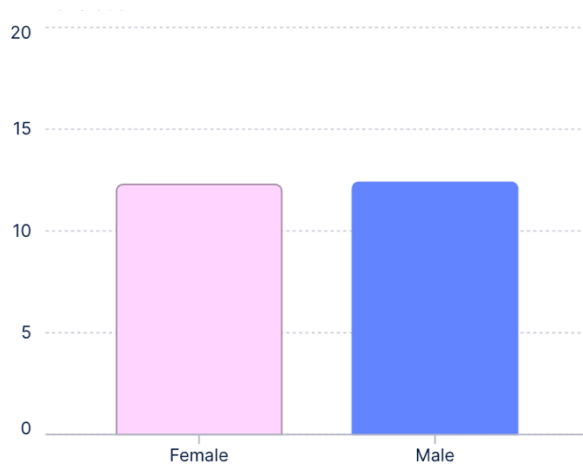


H3: Rejected



Informatics is not the highest group for post-defense corrections. Finance is higher.

Average final grade by gender



H5: Rejected



Grades are similar across gender groups, and no clear age-performance pattern.

ACTION PLAN & FINAL MESSAGE

Five actions to reduce avoidable process variation.



TARGETED SUPERVISOR DEVELOPMENT

Standardise feedback and reduce repeated correction loops.



PRE-SUBMISSION QUALITY CHECKLIST

Prevent avoidable structure, reference and plagiarism corrections.



EARLIER JURY FORMATION

Reduce jury revisions and rescheduling pressure.



REAL-TIME RISK MONITORING

Flag cases with correction loops, reschedules or final corrections.



INTERNATIONAL SCHEDULING SUPPORT

Support groups with higher rescheduling rates.



TITM does not need to redesign the whole thesis process.

The priority should be to reduce avoidable variation in the existing process.

Move from reactive problem-solving to proactive thesis process management.

Thank You Very Much For Listening

OPEN TO ANY QUESTIONS / DISCUSSIONS